

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

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Lab Objectives

- Understand basic algorithms using AI assistance
- Implement searching and sorting techniques
- Analyze time complexity
- Improve algorithm efficiency

Task 1: Linear Search

Python Code

```
def linear_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i] == key:  
            return i  
    return -1  
  
print(linear_search([10, 20, 30, 40], 30))
```

Explanation: Checks each element sequentially. Time complexity $O(n)$.

Task 2: Binary Search

Python Code

```
def binary_search(arr, key):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1

print(binary_search([10, 20, 30, 40], 30))
```

Explanation: Efficient search in sorted array. Time complexity $O(\log n)$.

Task 3: Bubble Sort

Python Code

```
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(n - i - 1):
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]
    return arr

print(bubble_sort([5, 3, 2, 4]))
```

Explanation: Simple sorting method. Time complexity $O(n^2)$.

Task 4: Selection Sort

Python Code

```
def selection_sort(arr):
    n = len(arr)
    for i in range(n):
        min_idx = i
        for j in range(i+1, n):
            if arr[j] < arr[min_idx]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr

print(selection_sort([5, 3, 2, 4]))
```

Explanation: Selects minimum element each iteration. Time complexity $O(n^2)$.

Task 5: Comparison

- Linear Search: $O(n)$
- Binary Search: $O(\log n)$
- Bubble Sort: $O(n^2)$
- Selection Sort: $O(n^2)$

Conclusion

AI helps generate and understand algorithms quickly.

Efficiency and correct implementation are essential for real-world applications.